

Question_1:

Code:

```
1  /*
2  Question_1: Find the missing value among M, D and X using  $M = -D * X$ .
3  */
4
5  #include <iostream>
6  #include <iomanip>
7  using namespace std;
8
9  int main() {
10     char var[3];
11     string inp[3];
12     double m = 0, d = 0, x = 0;
13
14     for (int i = 0; i < 3; i++) {
15         cin >> var[i] >> inp[i];
16         if (var[i] == 'M' && inp[i] != "?") m = stod(inp[i]);
17         if (var[i] == 'D' && inp[i] != "?") d = stod(inp[i]);
18         if (var[i] == 'X' && inp[i] != "?") x = stod(inp[i]);
19     }
20
21     for (int i = 0; i < 3; i++) {
22         if (inp[i] == "?") {
23             if (var[i] == 'M') m = -d * x;
24             if (var[i] == 'D') d = -m / x;
25             if (var[i] == 'X') x = -m / d;
26
27             cout << char(tolower(var[i])) << " "
28                 << fixed << setprecision(2)
29                 << (var[i] == 'M' ? m : var[i] == 'D' ? d : x);
30         }
31     }
32     return 0;
33 }
```

Output:

Test Case_1:

```
M 14.00  
D -4.80  
X ?  
x 2.92  
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
M 12.00  
D -5.80  
X ?  
x 2.07  
PS C:\Users\Prabin Yadav\
```